

A New Channel Selection Method using Autoencoder for Motor Imagery based Brain Computer Interface

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Abstract—To improve the spatial resolution of the electroencephalogram (EEG) signal, it is conventional to use a large number of scalp electrode while recording oscillatory rhythms in motor imagery based brain-computer interface(MI-BCI). However, this increases the dimension of data and might fail to generalize and thus over-fit. Therefore, it is required to reduce the dimension of input data in an optimal way. In this paper, we propose a method using an artificial neural network to reduce the dimension of EEG for MI-BCI. We train an under-complete sparse autoencoder neural network for each subject separately to encode the EEG data optimally. The optimally encoded EEG trials are then used by the Filter Bank Common Spatial Pattern (FBCSP) method to decode the imagined motor movement. In similar lines, autoencoder was trained on subject independent data. We achieved improved motor imagery classification accuracy when the dimension of the data was almost reduced by half compared to the state-of-art FBCSP. The performance of the proposed method is also compared with Sparse Common Spatial Pattern (SCSP) based channel selection method. The average classification accuracy obtained for 10 subjects is $74.3 \pm 8.06\%$ with only 13 encoded channels. Also, for the autoencoder trained to be subject independent we obtained an average classification accuracy of $66.64 \pm 3.93\%$ with only 11 encoded channels after cross-validation. The study extends the use of autoencoder neural networks in motor imagery based brain-computer interface and shows significant improvement in performance with reduced data dimension.

I. INTRODUCTION

Brain-Computer Interface (BCI) technology aims to provide an alternative communication and control means for patients suffering from neuromuscular disorder. In Motor Imagery BCI (MI-BCI), brain signals are recorded from the motor cortex region of the brain when the subject is imagining the movement or action of a particular motor organ (e.g., right hand). The imagination of the motor task blocks or desynchronizes the ongoing Mu rhythms (8-12Hz) in the contralateral part followed by an increase or synchronization of Alpha rhythms (12-18Hz) in ipsilateral part of brain's motor cortex. This phenomenon is termed as Event Related Desynchronization/Synchronization (ERD/S) respectively [1]. Traditionally, in an electroencephalogram (EEG) based MI-BCI system a large number of electrodes are placed all over the scalp to capture the oscillatory rhythms which have discriminative information during motor imagination task [2].

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To improve the spatial resolution of EEG, a large number of electrodes are used. This will increase the computational complexity as well as, the set up time for signal acquisition and will result in over-fitting of the data [3]. These issues are tackled by selecting fewer number of EEG channels(electrodes) that have more discriminative information, based on the prior knowledge of neurophysiology associated with motor imagery task.

The channel selection or dimension reduction methods can be filtering-based or search-based. In filtering-based method, certain measures like Euclidean distance or statistical measures are used to rank the channels and the top ranked few channels are used for further analysis. In [4], Fisher's ratio was calculated using two motor imagery class data (say, left vs right hand) for each channel and ranking was done based on Fisher's ratio. The work in [5] uses mutual information, [6] uses correlation on spectral entropy, and [7] proposed an iterative method based on difference between the band power during ERD and ERS activity to rank the channels. The search-based methods for reducing data dimension are classifier dependent, in which a subset of original data is obtained using search algorithms such as forward or backward sequential search in order to achieve better generalization or avoid over-fit [8]. The variant of Common Spatial Pattern (CSP) known as sparse CSP [9] ranks the channels after solving sparse optimization problem. The recursive feature elimination to remove task irrelevant feature in Support Vector Machine (SVM) proposed in [10], is a classifier dependent method Genetic Algorithm (GA) is used to arrive at the feature subset using Rayleigh co-efficient in [11]. The top ranked channels selected after ranking in filtering-based method will be mainly helpful in reducing the setup time. However, a generic BCI system should be capable of detecting not only oscillatory rhythms from motor cortex but also different states of the human brain [12]. For example, both attention level and motor related oscillatory rhythms need to be analyzed in MI-BCI. Further, the filtering methods take multiple steps to assign ranks to the channels and often do not consider the interaction or correlation of all different channel combination while ranking. The search-based methods are computationally intensive, classifier dependent and give sub-optimal results.

In this paper, we propose a EEG data dimension reduction method to eliminate the non-discriminative information from discriminative information, by encoding the data in an optimal way using sparse autoencoder. The autoencoder is trained using the data from a few trials during MI-BCI task for each subject. The trained encoder is then used to

reduce the dimension of EEG trials recorded for each subject. The method is compared with state-of-art Filter Bank CSP (FBCSP) [13] and Sparse CSP (SCSP) [9]. The performance improvement achieved with reduced dimension is discussed. In similar lines, we trained autoencoder using data from multiple subjects and cross-validated the performance achieved in terms of classification accuracy on data with reduced dimension.

The rest of the paper presents the experimental setup to acquire data in Section II, the proposed framework for channel selection using autoencoder in Section III, results and the comparison with state-of-art method in Section IV and finally we conclude in Section V.

II. EXPERIMENTAL SETUP

In this section, we present the experimental protocol designed for EEG data acquisition and signal preprocessing steps applied to enhance the desired EEG signal characteristics.

A. Data Recording

An experiment was designed to study the oscillatory rhythms during motor imagination tasks in a BCI protocol. Ten subjects (3 female and 7 male) aged $26.6 \pm 5.5\%$ years volunteered to participate in this study. This research was approved by the Institutional Review Board of Nanyang Technological University (NTU), Singapore (IRD - 2016-07-019) and data acquisition was conducted in NTU.

EEG was recorded from 31 electrodes placed on the scalp according to the international 10-20 standard with a sampling rate of 500 Hz and an initial band-pass filter between 0.05 - 200 Hz by the acquisition system (actiCHamp, Brain Products). The electrode placement is as shown in Fig. 1.

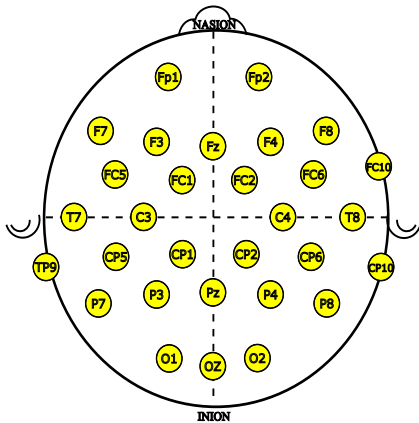


Fig. 1. Electrode placement.

During each trial, the subjects were asked to perform the imagination of clenching their right or their left hand according to the cue. A fixation cross appears at the center of the screen 2s before the visual cue. Then a green arrow pointing towards the left or right side of the screen appears at $t=0$ s indicating the motor action to be imagined. The subjects

were asked to imagine the motor movement until the green arrow disappears after 4s. The fixation cross reappears at the center of the screen again for 2s (relax period) and the trial ends (Fig. 2). The session had 50 such trials for each class and thus 100 trials for bilateral motor imagination task. Additional information on the data is available in [14].

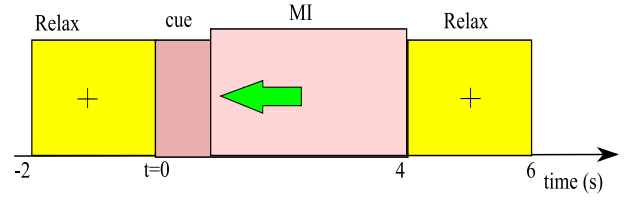


Fig. 2. Timing diagram of the visual cue presentation.

B. Data Extraction and Preprocessing

In this work, we consider the data recorded during motor imagination task. A smaller segment of motor imagery task data from each trial between $t=0.5$ s to $t=3.5$ s is considered for further analysis. The segmented raw EEG data belongs to $\mathcal{R}^{C \times N \times T}$, where C is the number of channels, N is the number of sample data points and T is the total number of trials.

Next, we pre-process the EEG signal using a band-pass IIR filter of order 4 with cut-off frequencies 0.5 Hz to 45 Hz. This step ensures the removal of slow varying artifacts, DC offset and unwanted high frequencies artifacts that may occur due to electromyogram (EMG). A notch filter is used to remove the power line noise.

III. PROPOSED CHANNEL SELECTION METHOD

In this section, we provide a brief review of Filter Bank Common Spatial Pattern Analysis (FBCSP) and present the architecture and training of proposed autoencoder neural network based method.

A. Filter Bank CSP

We describe the FBCSP method in brief. Common Spatial Pattern (CSP) analysis is a spatial filtering method used to compute the features of MI-BCI in optimal way by minimizing the variance of one class data while maximizing the variance of other class data at the same time. This can be considered as generalized eigen value problem and the simultaneous diagonalization of the covariance matrix gives the weight matrix [15]. If Y_i is the i^{th} class data ($i \in (1, 2)$) then co-variance matrix is given as $R_i = Y_i^T Y_i$ and the weight matrix W_{csp} required for the CSP transformation of the form $Z = W_{csp} Y$ is obtained from the solution of the generalized eigen value problem, $W_{csp} = eig(R_1, R_1 + R_2)$. The first and last m columns from W_{csp} will be used for transformation, assuming the weight vectors corresponding to the eigen values are in descending order. The log-variance of the transformed data is considered as the feature as given in (1). In FBCSP a set of filter banks are used and each pair of band-pass and spatial filter yields CSP feature corresponding to the frequency range.

$$v = \log\left(\frac{\text{diag}(Z^T Z)}{\text{tr}(Z^T Z)}\right) \quad (1)$$

B. Architecture of Autoencoder Neural Network

An autoencoder is a type of feedforward neural network which can be used to reduce the dimension of the data. The architecture of autoencoder is shown in Fig. 3 [16]. It consists of an input layer, a hidden layer, and an output layer. The number of neurons in the input and the output layer will be equal. The architecture considered has only one hidden layer and nodes in the hidden layer is varied between 2 to 20. The network can be considered to have two stages: an encoder function $h = g(W_1 X + b_1)$, which describes a code to represent input and a decoder function $\hat{X} = f(W_2 h + b_2)$, which reconstructs the input from the code. The parameters W_i and b_i are the weights and bias at the hidden layer ($i = 1$) and the output layer ($i = 2$) respectively. Our goal is to train the network to replicate the input as this will result in h which takes the useful properties of input. If we constrain the dimension (d) of h to be smaller than input dimension C then we get an under-complete autoencoder [16].

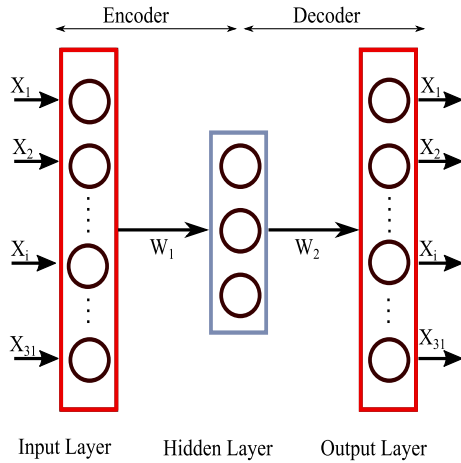


Fig. 3. Architecture of under-complete autoencoder.

As we are training the network to copy the input, the linear decoder function, $f(a) = a$ is used. The encoder function we have used in this work is sigmoid function $g(a) = \frac{1}{(1+e^{-a})}$ where a is a dummy variable. As we attempt to reduce the error between input X and the output $\hat{X}$, the loss function used is mean squared error. The loss function (2), is regularized to make the network more sparse and robust for missing or noisy data. The network is trained using scaled conjugate gradient algorithm.

$$L = \underbrace{(X - \hat{X})^T (X - \hat{X})}_{\text{MSE}} + \lambda * \underbrace{\|W\|_2}_{\text{L2}} + \beta * \underbrace{\Omega_{\text{sparse}}}_{\text{Sparsity}} \quad (2)$$

$$\text{where, } \Omega_{\text{sparse}} = \sum_{i=1}^k (KL(\rho \parallel \hat{\rho}_i))$$

The sparsity constraint in the loss function tries to ensure that a neuron is inactive most of the time. The sparsity regularization term will add a large value when the average activation $\hat{\rho}_i$ of neuron i and its desired value ρ is not close in value. Mathematically, it is the Kullback-Leibler divergence between $\hat{\rho}_i$ and ρ .

The FBCSP is implemented with $m=2$ and features are selected using Mutual Information based Best Individual Feature (MIBIF) and Support Vector Machine (SVM) classifier is used at the last stage as described in [13]. The schematic of the proposed method shown in Fig. 4. The first stage is dimensionality reduction, achieved using the trained autoencoder and the second stage is FBCSP, which extracts features and classifies the motor imagery data. The discriminative features extracted using FBCSP, the relevant features selected using MIBIF method and classified using SVM is the baseline method in this work.

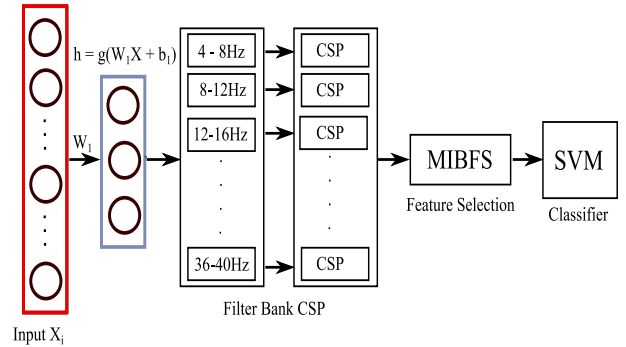


Fig. 4. Schematic of the proposed method.

IV. RESULTS

The autoencoder network proposed is trained using scaled conjugate gradient algorithm with L2 ($\lambda = 0.001$) and sparsity ($\beta = 1.5$) regularization. The section presents the results achieved by selecting the channels in two different ways. First, the autoencoder is trained with subject dependent data. Second, the autoencoder is trained with subject independent data. The results achieved are compared with state-of-art FBCSP method and SCSP method [9].

1) *Subject Dependent Autoencoder*: Each subject has 100 trials of MI data, out of which 25 trials were selected randomly to train the autoencoder. The trained autoencoder is used to encode data and thus reduce the dimension or channels. This is followed by the baseline method. 10-fold cross validation is performed in the classifier stage. The number of the nodes in the hidden layer represents the dimension of the encoded data and is varied from 2 to 20. The box plot (Fig. 5.) shows the average classification accuracy achieved for 10 subjects with different size of the hidden layer. The average classification accuracy obtained with 10 subjects using the proposed method is highest ($74.3 \pm 8.06\%$) with hidden layer size equals to 13. The classification accuracy for each subject with $h = 13$ is given in Table I.

TABLE I
PERFORMANCE OF EACH SUBJECT USING DIFFERENT METHODS

Subject ID	FBCSP [13]	Proposed method Subject Dependent(h=7)	Proposed method Subject Dependent (h=13)	Proposed method Subject Independent (h=11)
S-01	78%	78%	82%	74.2%
S-02	61%	67%	76%	70.8%
S-03	76%	68%	80%	71.6%
S-04	55%	80%	75%	65.8%
S-05	68%	69%	66%	63.6%
S-06	79%	74%	88%	68.6%
S-07	68%	67%	60%	59.4%
S-08	67%	64%	71%	71.4%
S-09	66%	66%	70%	59.4%
S-10	69%	69%	75%	61.6%
Average	68.7%	70.2%	74.3%	66.64%

The trained autoencoder is further analyzed by reconstructing the encoded data using the decoder part of the neural network (Fig. 3). The test mean square error (MSE) is calculated using 75 trials of EEG data from each subject that were not used for training the autoencoder network. The MSE for the number of nodes in the hidden layer equal to 7 and 13 is shown in Fig. 6. The mean MSE obtained for nine subjects(excluding subject, S-02) with hidden nodes equal to 7 is 4.68 and with hidden nodes equal to 13 is 2.18.

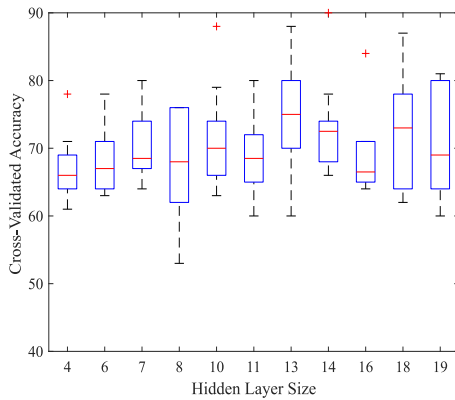


Fig. 5. The cross-validated classification accuracy obtained using different hidden size of the autoencoder trained for subject dependent data.

2) *Subject Independent Autoencoder*: The training of the autoencoder is done with same architecture as described in Fig. 4. 20 random trials from 8 subjects were used to obtain the weight and bias parameters of the autoencoder neural network. The encoder stage of the trained autoencoder is then used to encoded the EEG trials of all 10 subjects to a lower dimension. The reduced dimension data are then used by the baseline method. 10-fold cross validation is performed in the classifier stage. The nodes in the hidden layer is varied from 2 to 20. To cross-validate the proposed subject independent autoencoder method the above procedure is repeated for another 4 times with training data from different 8 subjects every time. The average classification accuracy achieved with different hidden layer size for each validation step is shown in Fig. 8. The leave two-out strategy shows that the performance of the proposed method is not

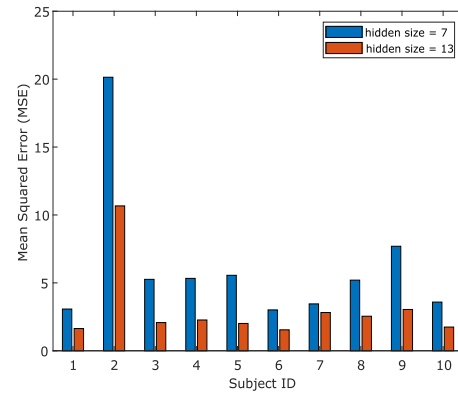


Fig. 6. The mean square error on test data for subject dependent autoencoder.

biased towards any particular set of training data.

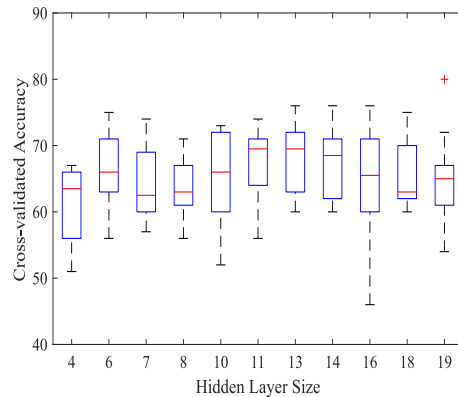


Fig. 7. The cross-validated classification accuracy obtained using different hidden size of the autoencoder trained for subject independent data.

The classification accuracy obtained from leave two-out cross validation is averaged. Thus obtained accuracy for different number of nodes in the hidden layer size is shown in the box plot (Fig. 7). The average classification accuracy obtained with 10 subjects using the proposed subject independent channel selection method is highest ($66.64 \pm 3.93\%$) with 11 nodes in the hidden layer, after the leave two-out

cross validation. The classification accuracy for each subject with $h = 11$ is given in Table. I

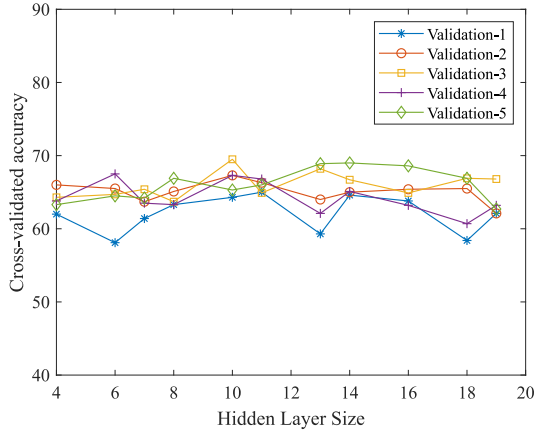


Fig. 8. The cross-validated accuracies for each validation step in autoencoder trained with subject independent data.

The EEG trials that were not used to train the autoencoder network (80 trials per subject) were concatenated to test the trained autoencoder for MSE analysis. The test MSE is large for a smaller number of nodes in the hidden layer and it decreases as the number of nodes are increased (Fig. 9.)

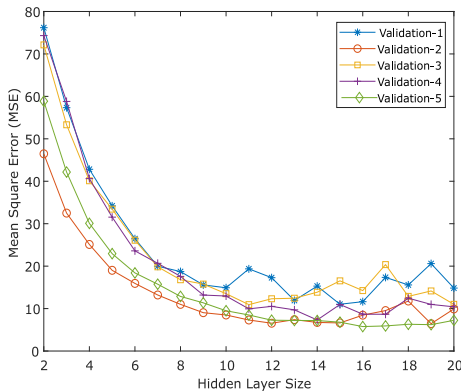


Fig. 9. The test mean square error of autoencoder trained using data from multiple subject.

3) *Comparison of results with FBCSP*: To compare the number of channels required to achieve certain classification accuracy, we implemented FBCSP [13] with 12 channels. The selected channels $\{F_z, FC_5, FC_1, FC_2, FC_6, C_3, C_4, CP_5, CP_1, CP_2, CP_6, P_z\}$ are from the motor cortex region. The average classification accuracy obtained using 12 channels for 10 subjects is $68.7 \pm 7.4\%$ whereas, the subject dependent autoencoder trained with hidden layer size (h) equals to 7 achieved average accuracy of $70.2 \pm 5.32\%$. The classification accuracy for each subject with $h = 7$ is given in Table I.

4) *Comparison of results with SCSP*: To compare the performance with SCSP, we rank the channels after solv-

TABLE II
COMPARISON BETWEEN SCSP AND PROPOSED METHOD

Subject-ID	SCSP [9]		Proposed Method	
	Accuracy	Channels	Accuracy	Channels
S-01	83%	18	84%	17
S-02	77%	11	76%	14
S-03	76%	19	80%	14
S-04	74%	19	81%	18
S-05	79%	11	73%	15
S-06	92%	20	90%	15
S-07	75%	20	75%	10
S-08	75%	9	70%	10
S-09	73%	20	71%	10
S-10	79%	19	75%	14

ing the sparse optimization problem [9]. In the proposed autoencoder method, we varied the hidden size between 2 to 20 and therefore, comparison with SCSP was done using 19 top ranked channels. The baseline method is applied on the channels selected using SCSP to obtain the classification accuracy. The number of channels required to obtain maximum accuracy for each subject with SCSP and the proposed subject dependent autoencoder method is given in Table II. The classification accuracies obtained for most subjects are identical for both the methods. For 7 subjects (S-01, S-03, S-04, S-06, S-07, S-09, S-10), the number of channels required to achieve maximum classification accuracy in proposed method was significantly less than that of SCSP [9]. For subjects S-08, S-02 and S-05, the proposed method needs 1, 3 and 4 channels respectively more than SCSP [9]. Overall, the proposed method offers almost identical classification accuracy using fewer number of channels compared to SCSP [9]. Further, SCSP formulation is nonlinear optimization problem which ranks the channel during offline analysis. The proposed method finds the optimal combination of channels that can characterize the underlying activity.

V. CONCLUSIONS

The objective of the work was to reduce the dimension of the EEG data to avoid overfit and better generalize the EEG data with optimal number of channels. The proposed sparse under-complete autoencoder based channel selection method shows that the classification accuracy achieved with different nodes in the hidden layer was always much above chance level for all the subjects. The number of channels was almost reduced to half in achieving slightly better classification accuracy compared to state-of-art FBCSP. The comparison with another spare optimization channel selection method (SCSP) shows that the number of channels were reduced in the proposed method. The classification accuracy of the subject independent channel selection using autoencoder is above chance level for all the subjects. The leave two-out cross validation of the proposed method proves the efficiency of the method.

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